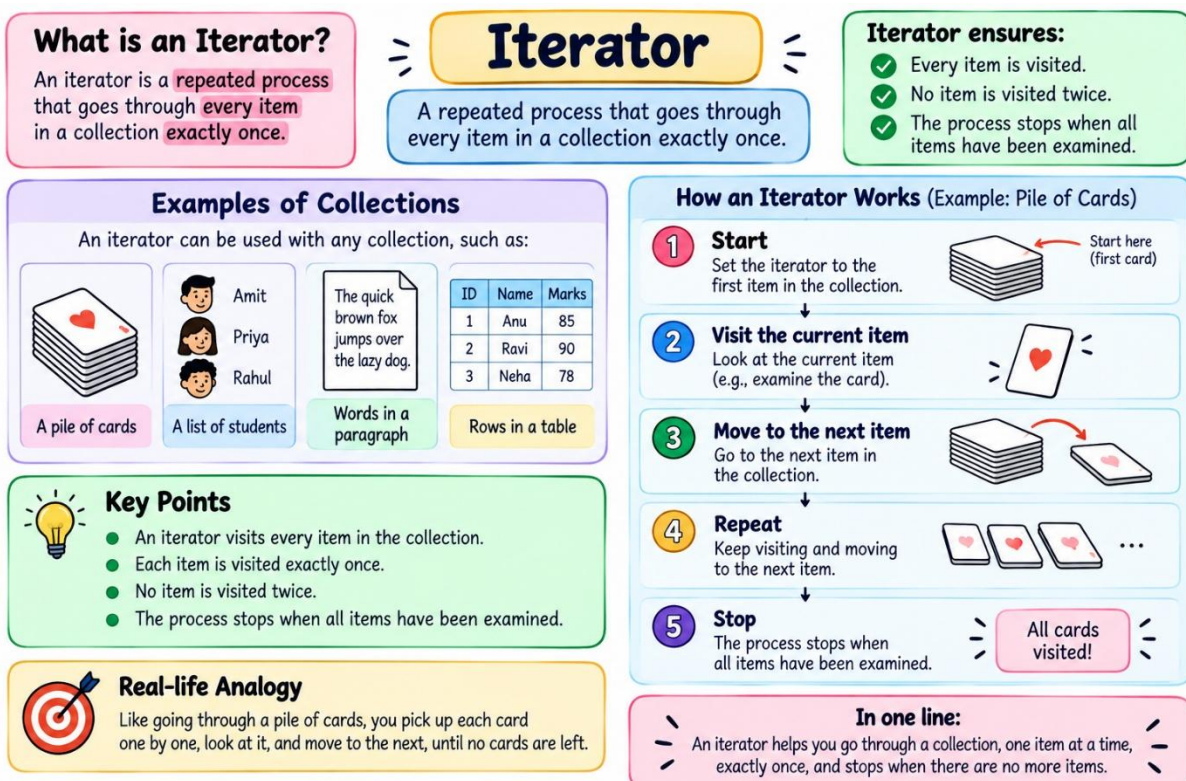


Iterator : The pattern of doing something repetitively is called iterator.

Property of Iterator is “Every item is visited and no item is visited twice”.

Description/parts of Iterator:

1. Initialization step
2. Continue or exit (condition check)
3. Repeat step (increment/decrement)

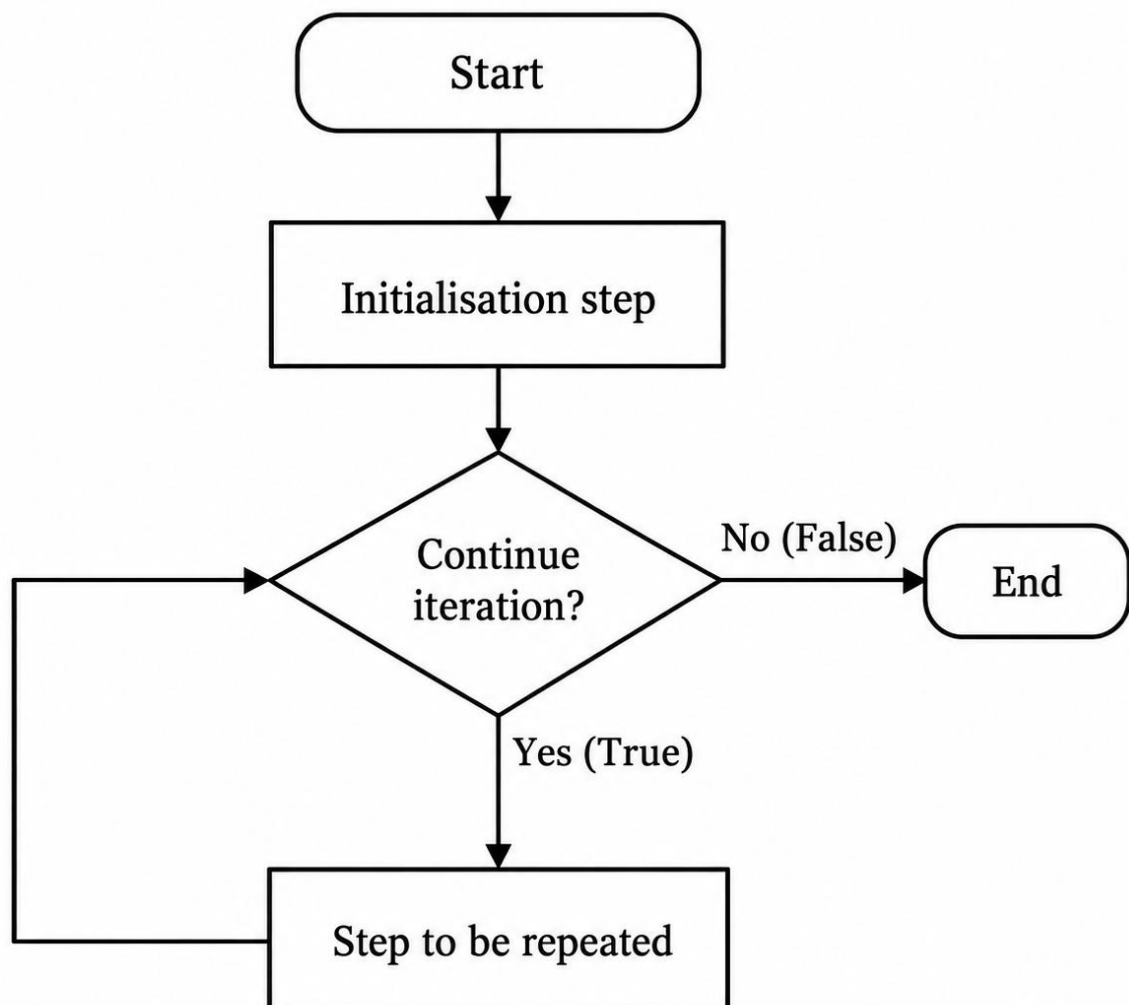


Dataset vs Data ElementThink of it like a **student information system**:

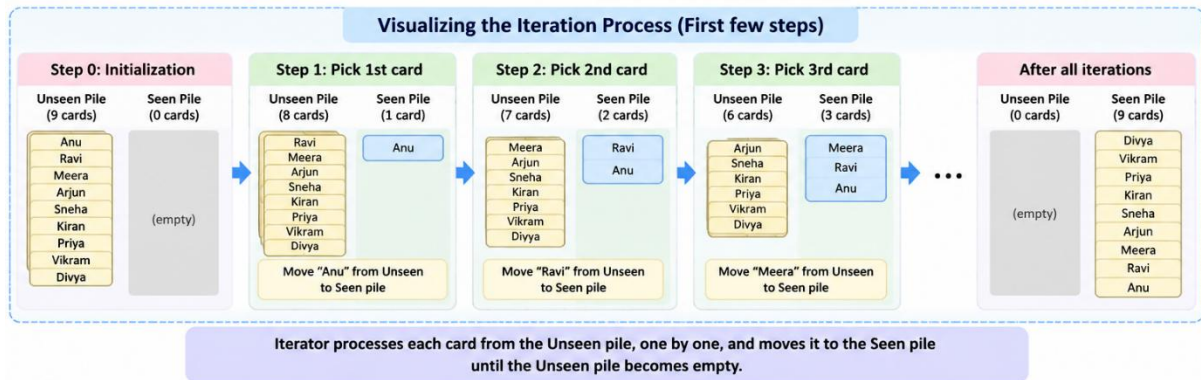
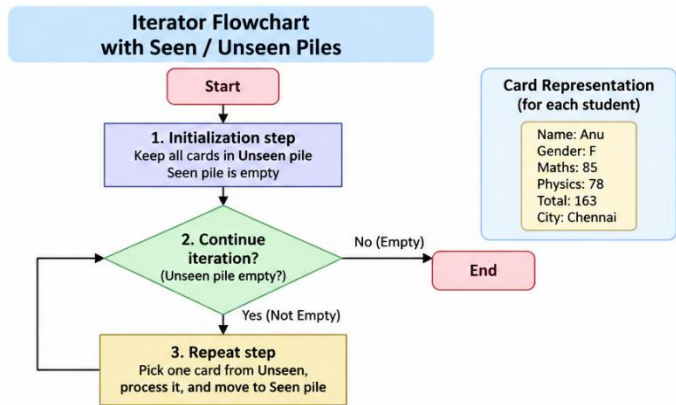
Concept	Meaning	Example
Data Element	A single piece of information	Student Name = "Ravi"
Dataset	A collection of related data elements/records	Complete student table

	Maths		Physics		Total	
1	Name	Gender	Marks	Marks	Marks	City
2	Anu	F	85	78	163	Chennai
3	Ravi	M	92	88	180	Hyderabad
4	Meera	F	76	84	160	Bengaluru
5	Arjun	M	68	73	141	Chennai
6	Sneha	F	95	91	186	Visakhapatnam
7	Kiran	M	81	79	160	Vijayawada
8	Priya	F	89	94	183	Hyderabad
9	Vikram	M	74	68	142	Bengaluru
10	Divya	F	98	96	194	Chennai
11	Rahul	M	87	82	169	Visakhapatnam
12	Kavya	F	72	77	149	Vijayawada
13	Suresh	M	63	71	134	Chennai
14	Nisha	F	86	89	175	Bengaluru
15	Amit	M	91	93	184	Hyderabad
16	Pooja	F	79	83	162	Visakhapatnam
17	Manoj	M	70	66	136	Vijayawada
18	Lakshmi	F	93	87	180	Chennai
19	Vijay	M	84	90	174	Bengaluru
20	Isha	F	67	75	142	Hyderabad
21	Harish	M	96	89	185	Visakhapatnam

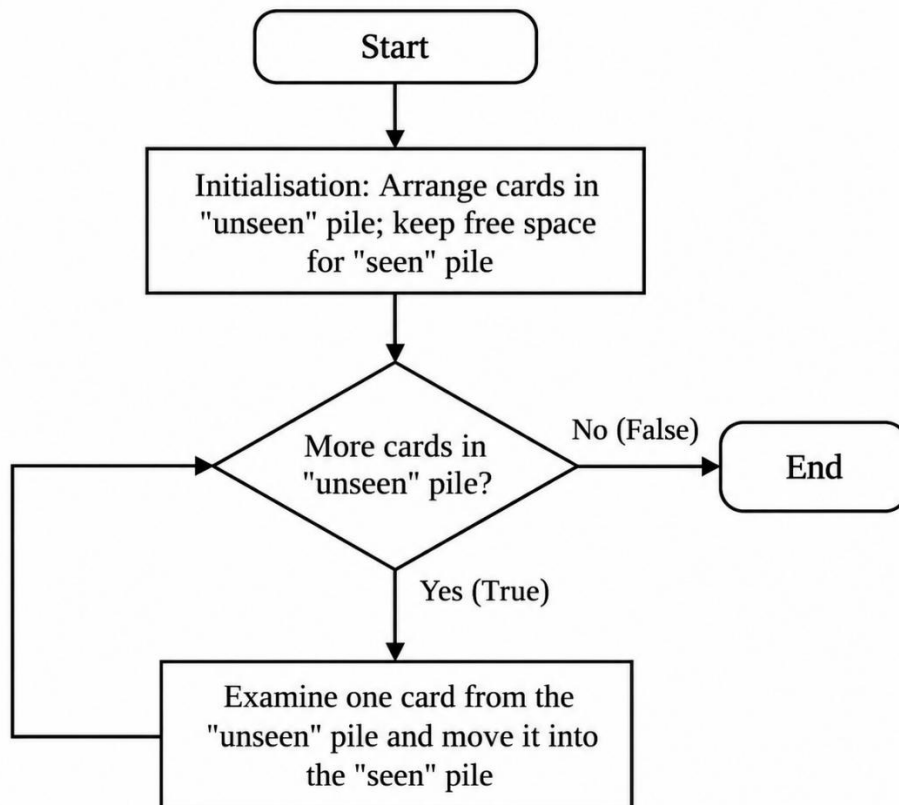
Generic Iterator Flowchart



Student Dataset (Cards)					
Name	Gender	Maths	Physics	Total	City
Anu	F	85	78	163	Chennai
Ravi	M	92	88	180	Hyderabad
Meera	F	76	84	160	Bengaluru
Arjun	M	68	73	141	Chennai
Sneha	F	95	91	186	Visakhapatnam
Kiran	M	81	79	160	Vijayawada
Priya	F	89	94	183	Hyderabad
Vikram	M	74	68	142	Bengaluru
Divya	F	98	96	194	Chennai



Iterator flowchart for cards



Store
Use
Change
Reuse

Variables in Programming

A variable is like a labeled box to store data in memory

Think of a variable as a container with a name, where you can put a value, change it, and use it later.

1. What is a Variable?
 A variable is a named memory location used to store data. The value can change during program execution.

Here, 'age' is a variable and 20 is its value.

2. Key Points

- 1 A variable has a name.
- 2 A variable stores a value.
- 3 The value can change.
- 4 It has a data type (int, float, char, etc.).
- 5 We can use the variable value later in the program.

3. Variable Declaration and Initialization

```
int age = 20;
```

Data type
(tells what type of data)

Variable name

Initial value
(stored in memory)

4. How it Works?

```

      Declare a variable
      ↓
      Initialize with a value
      ↓
      Use the variable
      ↓
      Change the value (if needed)
      ↓
      Use it again
    
```

5. Example (Java)

```

      int age = 20;
      String name = "Anu";
      double marks = 85.5;
      boolean isPassed = true;
      // we can change the value
      age = 21;
    
```

6. Variable in Action

7. Common Data Types

Data Type	Example	What it stores
int	10	Whole numbers
double	85.5	Decimal numbers
char	'A'	Single character
String	"Anu"	Text (characters)
boolean	true / false	True or false value

8. Real-life Analogy

- Box label → variable name
- Item inside → value
- You can replace the item with a new one → value can change
- You can use the item later → reuse

9. Rules for Naming Variables

- ✓ Can start with a letter, _ (underscore) or \$
- ✓ Can contain letters, digits, _ or \$
- ✓ Cannot start with a digit
- ✓ No spaces (use studentName, not student name)
- ✓ Case-sensitive (age and Age are different)
- ✗ Cannot use reserved keywords (like int, class, etc.)

10. Example Program Output

```

      int marks = 85;
      System.out.println("Marks: " + marks);
      marks = 90;
      System.out.println("Updated Marks: " + marks);
    
```

Output:
 Marks: 85
 Updated Marks: 90

Same variable, different value at different time!

"A small name, a big power — Variables make programs dynamic!"

Generic flowchart for iterator with variables

